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DLM assignment 3.

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Q.1) what are counters?

Sol^n:- A counter is a device which stores the no. of times a particular event or process has ocured, often in relationship to a clock signal. counters are used in signal electronics for counting purpose, they can count specific event happening in the circuit. for example, in up counter a counter increases count for energy runing edge of clock. Not only counting, a counter increases count for every rising edge of clock can follow the certain sequence based on our design line any random sequence 0,1,2,3..

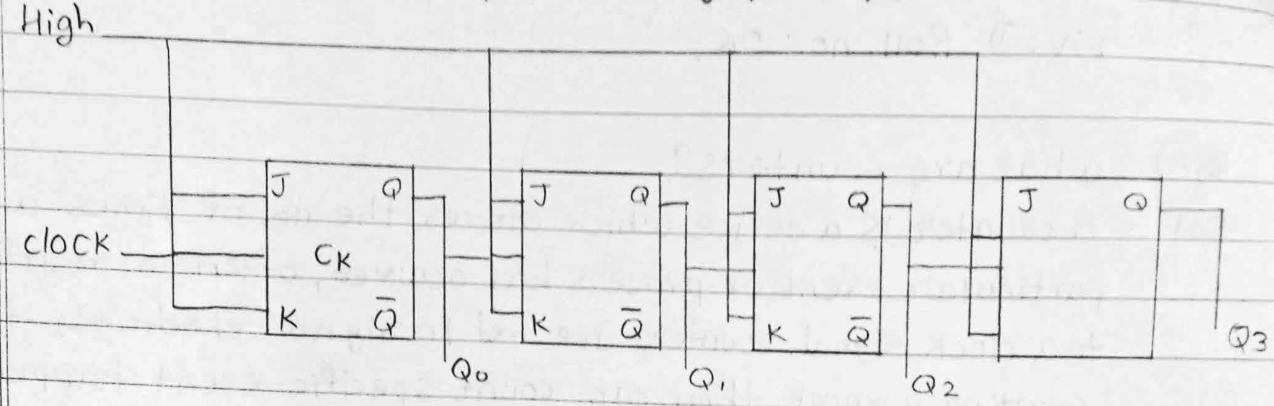
They can also be designed using the help of flip-flops. They are used as frequency dividers where one frequency of given pure waveform is divided. counters are sequential circuit one count me number of pulses can be either in binary code or BCD form. The main properties of a counter are timing, sequencing and counting.

counter works in 2 modes
- up counter
- down counter

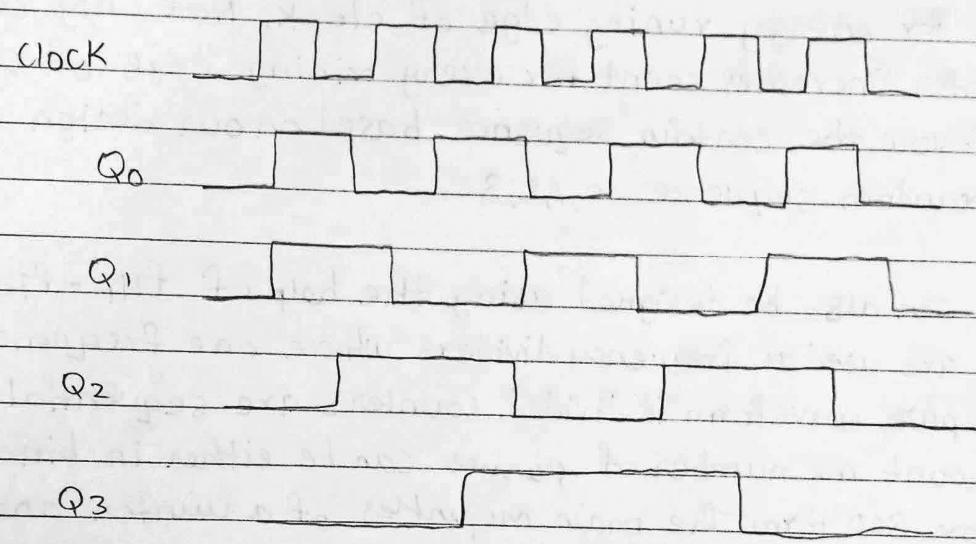
counters are broadly divided into 2 categories

- Asynchronous counter:-

In asynchronous counter we don't need universal clock, only first flop is driven by main clock and clock input of rest of the following flip flop is driven by output of previous flip-flops.



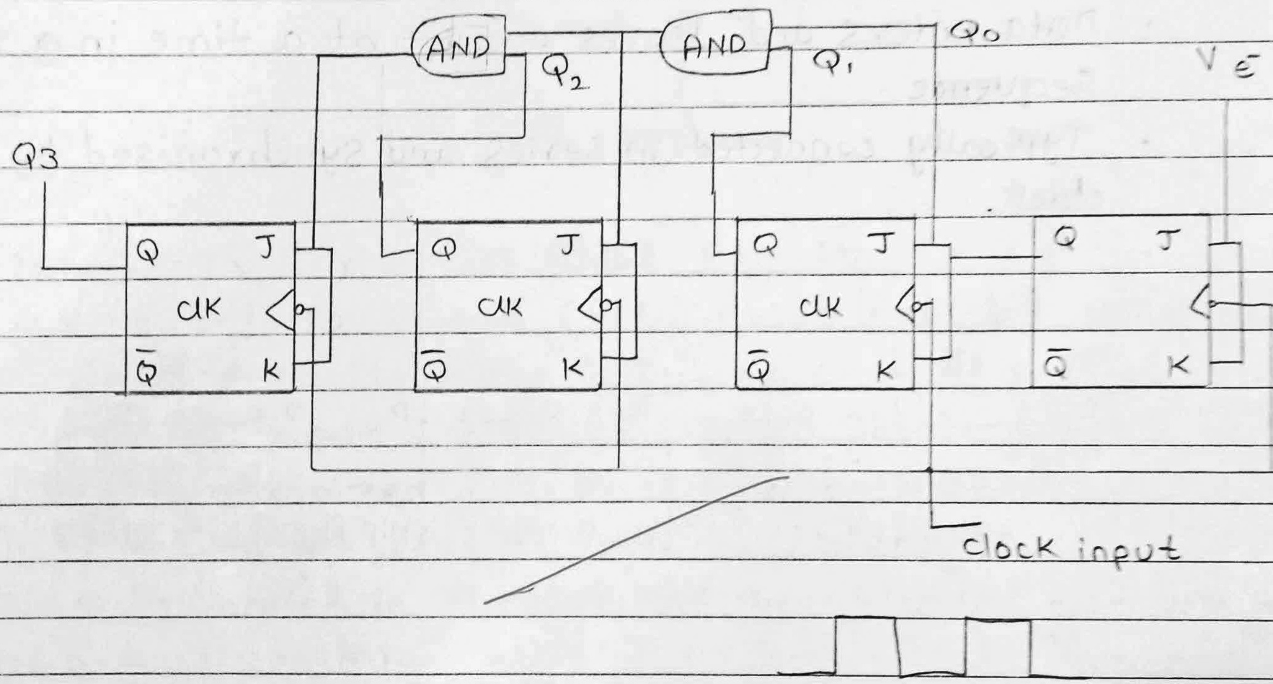
(a) ASynchronous counter.



(b) Timing diagram

It is evident from timing diagram that so is changing as soon as the rising edge of clock pulse is encountered. Q_1 is changing when rising edge of Q_0 is encountered and so on. In this way ripples are generated through Q_0, Q_1, Q_2, Q_3 hence it is also called RIPPLE counter and u rial counter. A ripple counter is a cascaded arrangement of flip flops where the output of one flipflop drives the clock input of the following flip flop.

Synchronous counter:
Unlike the asynchronous counter, synchronous counter has one global clock which drives each flip flop so output changes in parallel. The one advantage of synchronous counter over a synchronous counter is, it can operate on higher frequency than asynchronous counter as it does not have propogation delay because the same clock is given, to each flip-flop drives the clock input of one of the following flip-flop
It is also called parallel counter.



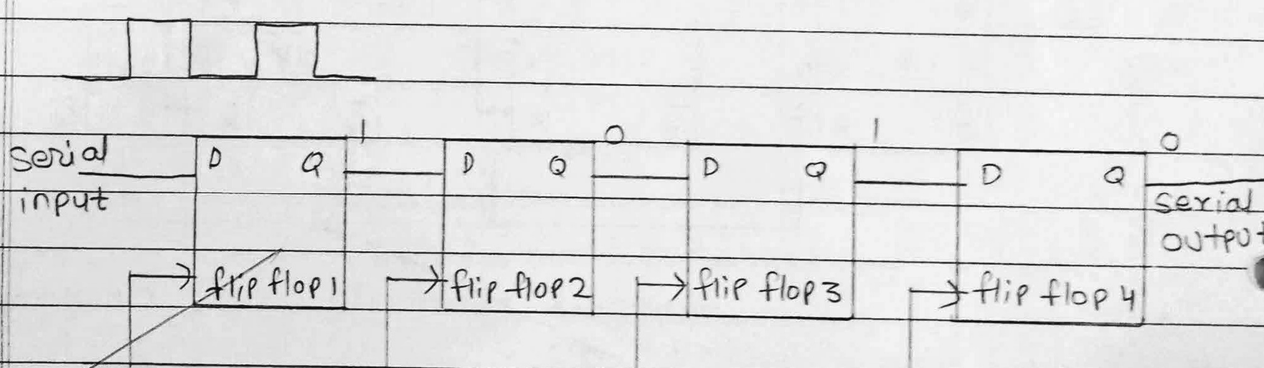
Q.2) what are shift registers?
solⁿ:- A shift register is a group of flip-flops connected in series to store and shift multiple bits of data either left or right with clock pulses.

- The registers which will shift the bits to the left are called "shift left registers".

• The registers which will shift the bits to the right are called "shift right registers".

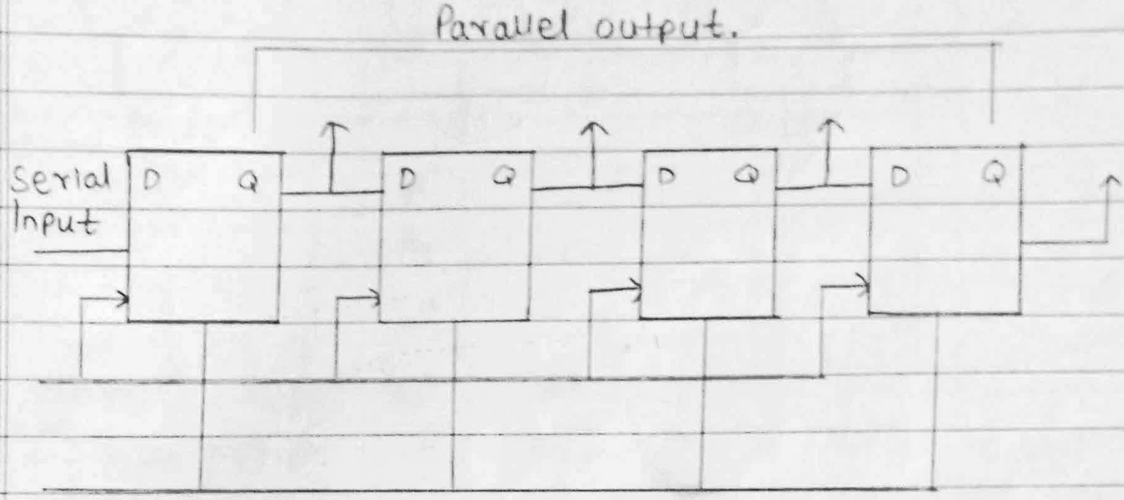
⊙ Serial-In Serial-out Registers (SISO).
A serial-In Serial-out (SISO) shift register accepts data one bit at a time through a single input line and outputs data serially, using a series of flip-flops all triggered by the same clock. the main use of a SISO is to act as a delay element.

• Data enters and leaves one bit at a time in a serial sequence
• Typically connected in series and synchronised by a common clock.



⊙ Serial-In Parallel-out shift Register (SIPO).
A serial-In parallel-out (SIPO) shift register accepts data serially through a single input line and produces through a single input line and produces all ~~line~~ bits simultaneously as a parallel output, using multiple synchronized flip-flop. The main use of SIPO register is to convert serial data into parallel data.

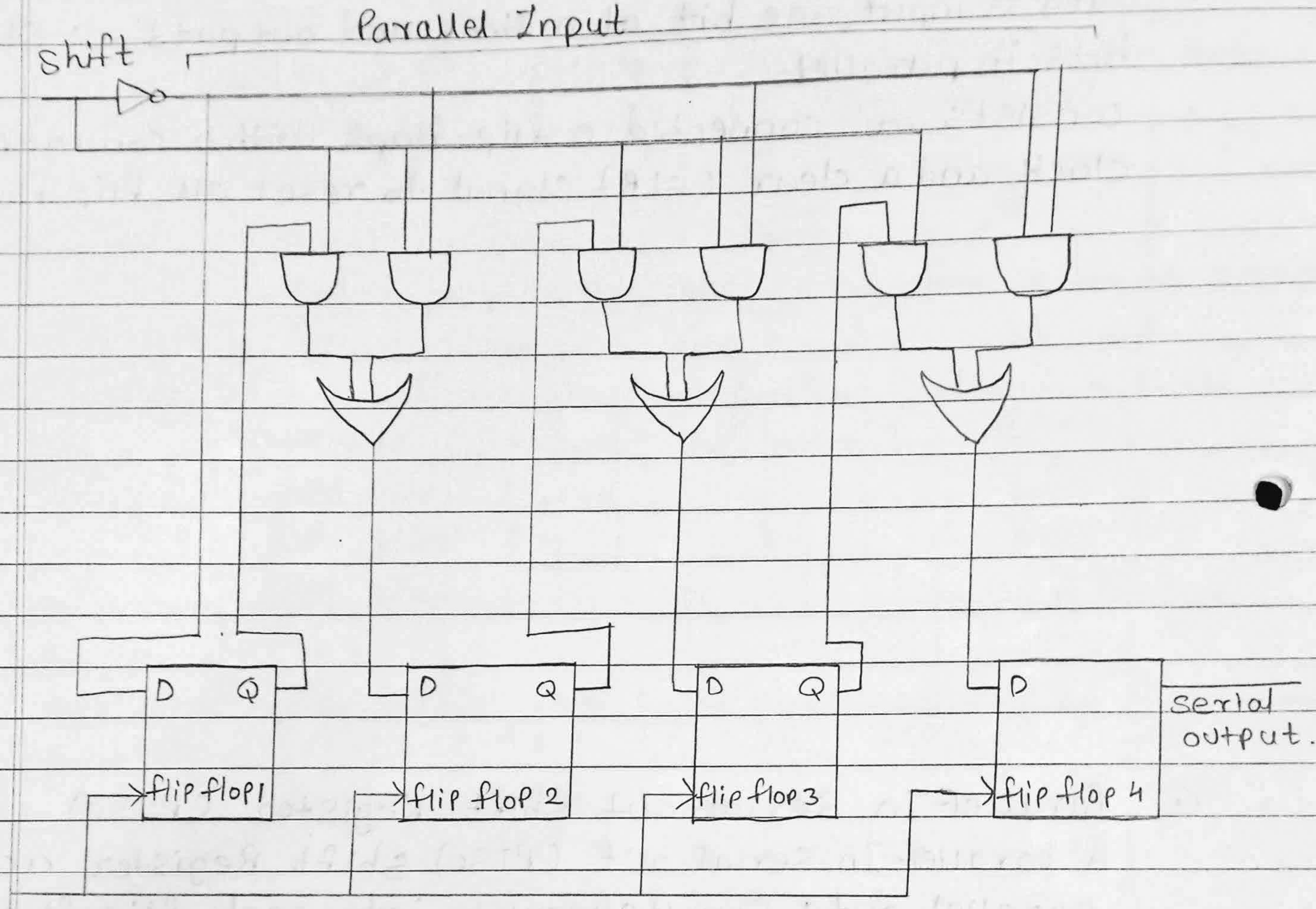
- Data is input one bit at a time and outputs all stored bits in parallel.
- consists of connected D flip-flops with a common clock and a clear (CLR) signal to reset all flip-flops.



⊙ Parallel-In Serial-out Shift Register (PISO)

A parallel-In Serial out (PISO) shift Register accepts parallel data simultaneously into each flip-flops and shifts in out serially through a single output line using synchronized flip-flops and multiplexars. It is used to convert parallel data to serial data.

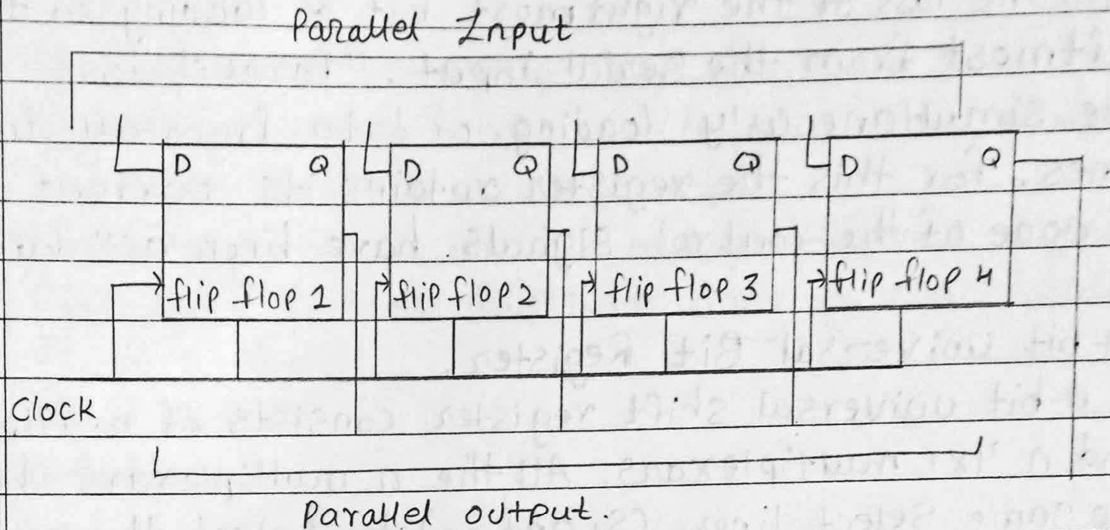
- Data is located in parallel and then shifted out one bit at a time serially.
- consists of D flip-flops with inputs controlled by multiplexers selecting between parallel data and privious flip-flop output, all cocked together.



⊙ Parallel-In Parallel-out shift Register (PIPO).

A parallel In parallel-out (PIPO) shift register accepts and output data simultaneously across all flip-flops without any serial shifting, using synchronized control signals. It is used as a temporary storage device and also acts as a delay element.

- Data is located and retrived in parallel from each flip-flop at the same time.
- Includes D flip-flops with no interconnections, controlled by common clock and clear (CLR) signals.



Q.3) What is universal shift Register?

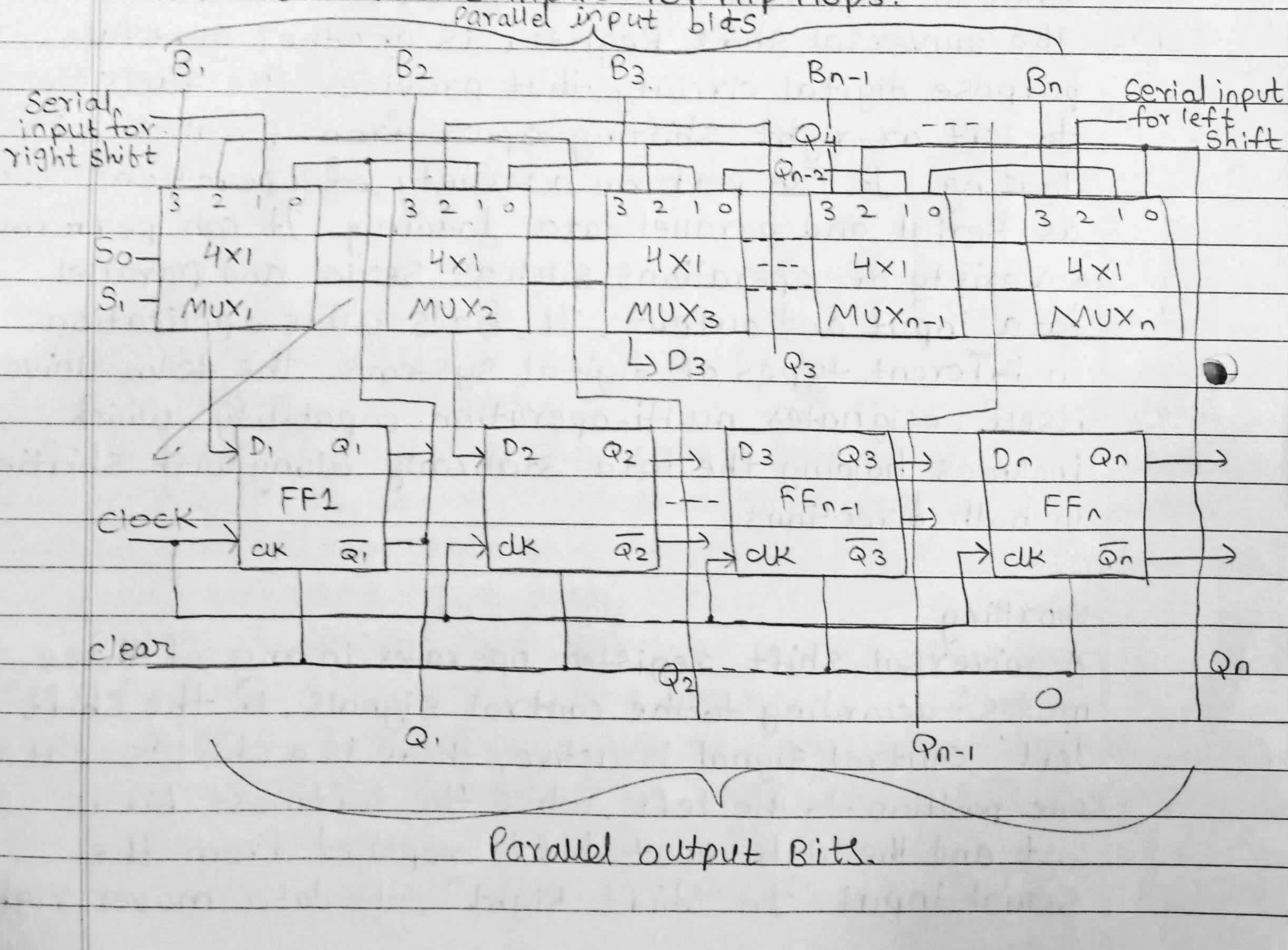
Sol^n:- The universal shift Register is another general-purpose digital circuity that provides the facility to left or right shifting apart from parallel-data loading. It can perform a variety of operations such as serial and parallel data loading. It can perform a variety of operations such as serial and parallel data input and output. It finds wide applications in different types of digital systems. the term universal itself designates multi-operation capability, which includes holding the data statically along with shifting in both directions.

Working.

A universal shift register operates in any of three modes: according to the control signals. If the "shift left" control signal is active, there is a shift is data one position to the left, while the leftmost bit is lost and the rightmost bit is replaced from the serial input. In "Shift Right", the data moves right with the loss of the rightmost bit & loading of the leftmost from the serial input. "Parallel load" allows the simultaneously loading of data from all input lines. for this the register retains its current data if none of the control signals have been activated.

(c) n-bit Universal Bit Register.

A n-bit universal shift register consists of n-flip flop and n 4x1 multiplexars. All the n multiplexars share the same select lines. (S_1 and S_0) to select the mode in which the shift register operates. The select input select the suitable input for flip-flops.



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Advantages of universal shift Register.

- Multifunctionality: left and Right shifts, parallel load, and static hold can be performed.
- flexibility: can be used in many different digital systems.
- Efficient: reduces the hardware to manipulate the data.

Disadvantages

- complexity: More complex and expensive as compared to the simple shift Registers.
- slower operations: Sometimes additional control logic makes them slower.
- High cost: Additional functionalities raise the cost of implementation.

Applications.

- Data Serializating / Deserializing: Data can be transferred from parallel to serial format and vice-versa.
- Delay circuits: The circuit that introduce the delay in digital signals are known as delay circuits.
- frequency divider: obtains lower frequency clock signals.
- counter and timers: counter and timers perform timing and counting operations.

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Kulkarni